A Brazilian Sign Language Video Database for Automatic Recognition

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***Abstract*— Communication is a basic human necessity, but the deaf community is still facing some barriers. In Brazil, there are about 9.7 million people with some degree of hearing impairment but most Brazilians do not know the Brazilian Sign Language (Libras). Wearable gadgets like gloves or bracelets with sensors are a good tool to help in communication, but these solutions are pricey or intrusive. On the other hand, computer vision solutions are cheaper but challenging mainly due to the absence of reliable video databases. In this work, we propose a video database compounded by different signs, with various backgrounds, and different speakers. We also developed a system to perform computer vision-based sign recognition using the k-nearest neighbors (K-NN) and random forest (RF). In our tests, we achieved an average accuracy of 65.81% in multiclass classification.**

***Index Terms*— Sign Language, LIBRAS, Computer Vision, Gait Energy Image, Image Processing.**

I. INTRODUCTION

Communication is a human necessity and the spoken language is a more common way to establish a dialogue. Nevertheless, the communication of deaf people is based on corporal gestures, facial expressions, and the local culture. According to the most recent Brazilian Census [1], 5% of the Brazilian population has some hearing loss and only 15% of these are fluent in Brazilian Portuguese. Most of the deaf population has as the first language the Brazilian Sign Language (also known as Libras) and the Portuguese as a second language. Despite this, the deaf population has a lot of barriers to use written communication. They are very different in syntax and semantics [2].

The Libras brings together the position of hands, the facial expression, and the gestures performed by the speaker to translate a word or a sentence. Besides, the same position of hands and facial expression, but with slightly different hands trajectories may have various different interpretations [3]. Therefore, to create a dictionary or an interface to translate those symbols is a hard task.

To recognize gestures, the literature presented two main approaches. The first one uses some devices as colored gloves or sensors to easily track the movements and that is known as "visual-based gesture with glove–markers" (VBGwGM). The disadvantage of this approach is that the sensors can limit the movements turning the communication uncomfortable. On the other hand, the second way to extract the information of the gesture movements is using computational vision [4], [5], as know as "pure visual-based gesture" (PVBG). This type of approach is less intrusive using only cameras. However, the usage of computational vision could generate a lot of unrecognized gestures.

The sign languages are the main method used by hearing-impaired people. Each sign language has specific characteristics as grammatical rules and their structure. The base structure is compound by a collection of gestures, movements, postures and facial expressions that all together generate the equivalent of letters, words and sentences in spoken language [6]. The sign language follows the *patria* language of the users. Therefore, for each spoken language, we have a different sign language. In American English, we have the American Sign Language (ASL); in French, we have the French Signal Language. Portuguese is the 9th language in number of users in the world and Libras is its respective sign language. Interestingly, although Brazil and Portugal use the same spoken language called Portuguese, their respective sign languages, Libras and Portuguese Sign Language (in Portuguese Língua Gestual Portuguesa, LPG) are completely different. The same occurs between USA and UK with ASL and British Sign Language (BSL).

Despite the Libras popularity and usage, the literature presents few databases with gestures. The Brazilian Government, through the National Institute of Deaf Education (INES), maintains an online dictionary containing 3,853 lexical signs/items [7]. Despite the vast amount of signs, the dictionary has only one speaker, one background, and only one repetition of each word, making its usage for gesture recognition applications a hard task.

In this work, we will describe in detail the CEFET Libras

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Database, compounded by 547 videos with different gestures, hand positions, backgrounds and speakers without the aid of gloves or wearable tracking devices. This database may be used to train intelligent algorithms that can be implemented in robotic systems. We also proposed a computer-vision based method to recognize Libras signs in this database to serve as baseline. The method consists in a feature mapping based on the squared residual video, so the feature space can capture important details for Libras like expression, hand configuration and orientation as well as the motion. The classification of the words in the feature space can be performed by two classifiers, K-Nearest Neighbors (K-NN) or Random Forest (RF).

II. Related Works

In the literature, most of the works discourses about static symbol recognition. The posture is a static hand position configuration, no movement is considered. In this case, the database can be constructed using only images instead of videos.

Carneiro et al. [8] evidenced the static symbols topic tackling the the alphabet recognition for Libras using neural networks. The authors produced a database formed by 50 images for each of the 26 symbols made by 3 volunteers, making a total of 3,900 images. The gesture recognition for the alphabet achieves 90% of accuracy.

Souza et al. [9] used a database mixing static and dynamic signs to perform the translation from Libras to Brazilian Portuguese. The database is composed of 46 static symbols and 13 words corresponding to 13,810 images and 939 video sequences, respectively. Two separated classification structures were adopted: one for the symbol classification that uses Hidden Conditional Random Fields (HCRF), and the other for the gesture classification that uses Support Vector Machine (SVM). The work shows 80.19% of accuracy using a 10-fold validation strategy. It can be observed, as a primary conclusion, that the gestures strongly reduces the overall accuracy.

In order to exemplify the last comment, it will be described a classification framework for 61 static symbols presented in Costa et al. [10]. The hand segmentation was performed by using the resources of a Kinect and the video sequence. This work used Principal Components Analysis (PCA) and 1-NN reaching the best accuracy of 96.31%, which is a good result in despite of the increase in the number of symbols.

Using a different perspective, the newly published work by Rocha et al. [2] combined symbols to create gestures. Each symbol is represented by static sign or more than one static sign in the same video, which are recorded with Chroma Key as background. However the number of speakers and repetitions were not detailed. The whole database consists in 31 classes composed by letters, symbols and some words. The authors achieved 99.42% of accuracy using a model based on Convolutional Neural Network (CNN).

This similar occurrence of showing higher hit rate for static symbols recognition can also be observed for other sign languages. Considering the usage of ASL [11] for static symbols (alphabet and digits), the best accuracy result achieved 99.93% using a fine-tuned CNN.

Dias et al. [12] performed a gesture recognition applied to Libras. The proposed database is generated from the data of 4 speakers, two men and two women, repeating three times each gesture, making a total of 600 samples. Although no result is mentioned on the paper, the gesture recognition reported uses Hidden Markov Machine (HMM).

Yang et al. [13] recently published a work based on Leap Motion Controller (LMC) and Deep Learning to classify dynamic hand gestures. Although it was the aim of the paper, it shows clearly how difficult is to aggregate the dynamic signs to the recognition task and their impact into the systems performance. The position of each finger in the video is precisely obtained by using LMC and considering the usage of an external gadget, this approach can be classified as VBGwGM. The results for a mixed static (60%) and dynamic (40%) symbols database (namely as Handicraft-Gesture Database), for 10 classes, using 10-fold validation and two-layer Bidirectional Recurrent Neural Network was only 90% of accuracy. Despite the fact that each finger was marked in the videos.

A Libras database is presented in Escobedo et al. [14]. It comprises 37 signs performed by ten subjects: six women and four men, containing RGB-D and skeleton information collected using a Kinect device. The database is supposed to be public; however, as far as we know, no means to obtain is provided. Moreover, a detailed description of the database is not given, making a comparison with our database unfeasible. The work also proposed CNN-based models that are trained using multimodal information (RGB-D) through texture maps and hand configuration descriptors. Their best model achieved 87.02% of accuracy on the proposed database. As they use depth information in their method, a direct comparison with our results is unfair.

As a whole showed in this Section, we can conclude the classification of dynamic hand positioning is still a very challenging problem.

III. Database description

In this section, we present our database which consists of 547 video sequences and 24 entries (words or phrases). The video sequences have 640×480 pixels as resolution

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Fig. 1: Database screenshots showing different speakers and backgrounds. Each row represents video frames in which a volunteer is performing a gesture in a background. First row: volunteer ID04, in background F1, performing the word "scaffolding"; second row: volunteer ID03, in background F3, performing the word "ghost"; third row: volunteer ID04, in background F2, performing the word "miracle".

and 30 Hz as frame rate. The records were done in a controlled environment with three static backgrounds: simple (F1), intermediate (F2), and complex (F3). Each entry was performed by 20 non-deaf volunteers, 13 women and 7 men. This database is available under request[3].

The gesture sequence is performed in a video that describes a word entry as an acronym. The Table I shows all entries presented in our database and their respective acronym. It is possible to segregate the entries into two classes: the first one is with horizontal hands movement, and the other one, with vertical hands movements, as shown in Table II.

All the information about acronym, background, and volunteer identification is contained in each filename. For instance, the video named as `MIGR_F2_ID04.avi` is the video for acronym MIGR (word miracle), in background F2 by the volunteer ID04. Some screenshots are shown in Figure 1.

A. *Database statistics*

Following, we present some statistics of our database named CEFET LIBRAS Database. Figure 2 shows the duration (in seconds) distribution of the videos in the database. It is worth mentioning that the longest and the shortest videos of the database have 134 frames (5.37 seconds) and 14 frames (0.93 seconds), respectively. The average length of the videos is 50.05±22.18 frames (2.34±0.74 seconds).

Figure 3 shows a general information about the number of occurrences of the words in the database. The frequency of the words spoken by volunteers is depicted in Figures 3a and 3b showing the total number of frames for each word. We also analyze the distribution of the volunteers in the database, as shown in Figure 4, which shows the occurrence of each volunteer in the database. It is possible to notice that most of the speakers were recorded at least 25 times for different words.

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TABLE I: List of entries and their acronyms in the CEFET LIBRAS Database.

Entry	Acr.	Entry	Acr.	Entry	Acr.
appointment	COMP	hammer	MART	patience	PACN
baby	BEBE	impossible	IMPO	scaffolding	ADME
composing	REDC	lightning	RLAM	snow	NEVE
depression	DEPR	lover	AMAN	steam	VAPR
dice	DADO	miracle	MIGRE	surf	SURF
elastic	ELAS	nacionality	NACN	television	TELE
fast	VELZ	news	NOTC	trampoline	TRAM
ghost	FANT	orchestra	ORQT	wall	PARD

TABLE II: Dataset segregation according to hand movements (horizontal or vertical).

Movement	**Entries (acronyms)**
Horizontal	lover (AMAN), baby (BEBE), appointment (COMP), dice (DADO), elastic (ELAS), impossible (IMPO), miracle (MIGRE), news (NOTC), orchestra (ORQT), surf (SURF), television (TELE) and fast (VELZ).
Vertical	scaffolding (ADME), depression (DEPR), ghost (FANT), hammer(MART), nacionality (NACN), snow (NEVE), patience (PACN), wall (PARD), composing (REDC), lightning (RLAM), trampoline (TRAM) and steam (VAPR).

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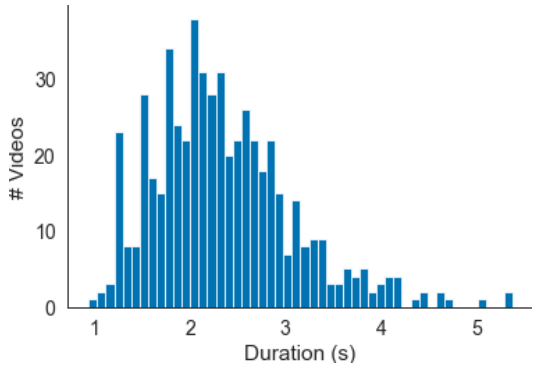


Fig. 2: Histogram of videos duration in our database.

IV. Automatic Gestures Recognition

In this section, we describe a proposed method to recognize Libras signs in the described database to validate our database and to serve as a baseline for future works that will use our database. It is important to highlight that each video (640×480) was resized to $q \times q$ ($q \leq 480$) using bicubic interpolation before each feature extraction method. The feature extraction, detailed in Section IV-A consists in mapping the video frames into a feature space that we call S-Residue. After that, we perform the classification as described in Section IV-B. This pipeline is illustrated as a block diagram in Figure 5.

A. Feature Extraction using residues

In our previous work [15], performed feature space mapping through the absolute residue according to:

$$R = \sum_{n=1}^{N} (I_{n-1} - I_n), \tag{1}$$

where I_n is the n-th frame in the video sequence. To obtain a specific size of features, the residual image R_n was split into cells and each cell is represented using the sum of their values. However, some movements were canceled using this approach and it has an impact over the accuracy of the system as can be seen in Section IV-C. In this work, we use a different approach to avoid this problem. We perform the feature space mapping, which we call S-Residue by summing the squared residual frames according to

$$S = \sum_{n=1}^{N} (I_{n-1} - I_n)^2. \tag{2}$$

B. Classification

In this work, we used multiclass K-NN ($K = 1$) and random forest (RF) classifiers, in a Matlab implementation. We performed grid search for the number of trees optimization of the RF classifiers in the range $\{20, 50, 100, 150, 200, 300, 400\}$. All the other parameters were set to the default values.

TABLE III: Best mean and standard-deviation results across the folders.

Features	# words	Classifier	Hit rate (%)
Residue [15]	12	1-NN ($q = 7$)	75.00
	24	–	–
S-Residue	12	1-NN ($q = 20$)	74.33±5.46
		RF ($q = 5$, 300 trees)	87.23±6.45
S-Residue	24	1-NN ($q = 50$)	53.90±5.48
		RF ($q = 50$, 400 trees)	65.81±6.30

C. Results

We validate our results using k-fold ($k = 3$) cross validation so that the results in this section are the average across folders. We performed experiments varying q (Section IV) in the range $\{3, 5, 10, 20, 50, 100\}$. To better compare to [15], we ran our experiments in two versions of the database, one full and other with 12 words (following the same rationale applied in [15]).

Figure 6 shows the results when using K-NN as classifiers. One can note from the graphic that the accuracy drops significantly when the number words are increased. The K-NN is a simple classifier, which probably could not handle with the increase in the problem complexity when more words are taking into account. For the full database, the K-NN average accuracy is 53.90% for $q = 50$.

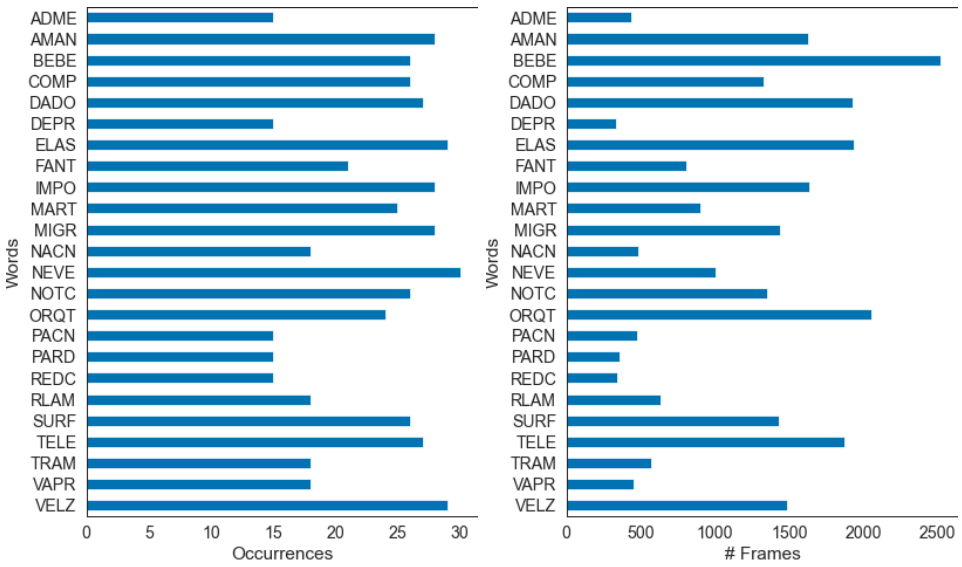
Figure 7 shows the performance when using RF, varying the number of trees. It is a much more powerful classifiers, so it could handle better with the full database. In this case, the best average accuracy is 65,81% when $q = 50$ and the number of trees is 400.

All results are summarized in Table III. The result in the first line was reported in [15]. When using K-NN, there is no significant difference between absolute and squared residue. In all cases, the accuracy drops when the number of word increases. However, when using S-Residue with RF as multiclass classifier, the accuracy remains satisfactory.

V. Conclusions

Technological tools for sign languages are very important to diminish the communication gap between the deaf community and the majority of the people. Tools using computer vision can be very powerful since they can be less intrusive. However, computer vision solutions are extremely dependent on the available data. In Brazil, there are about 9.7 million people with some degree of hearing impairment, but there is not a large reliable database for computer vision

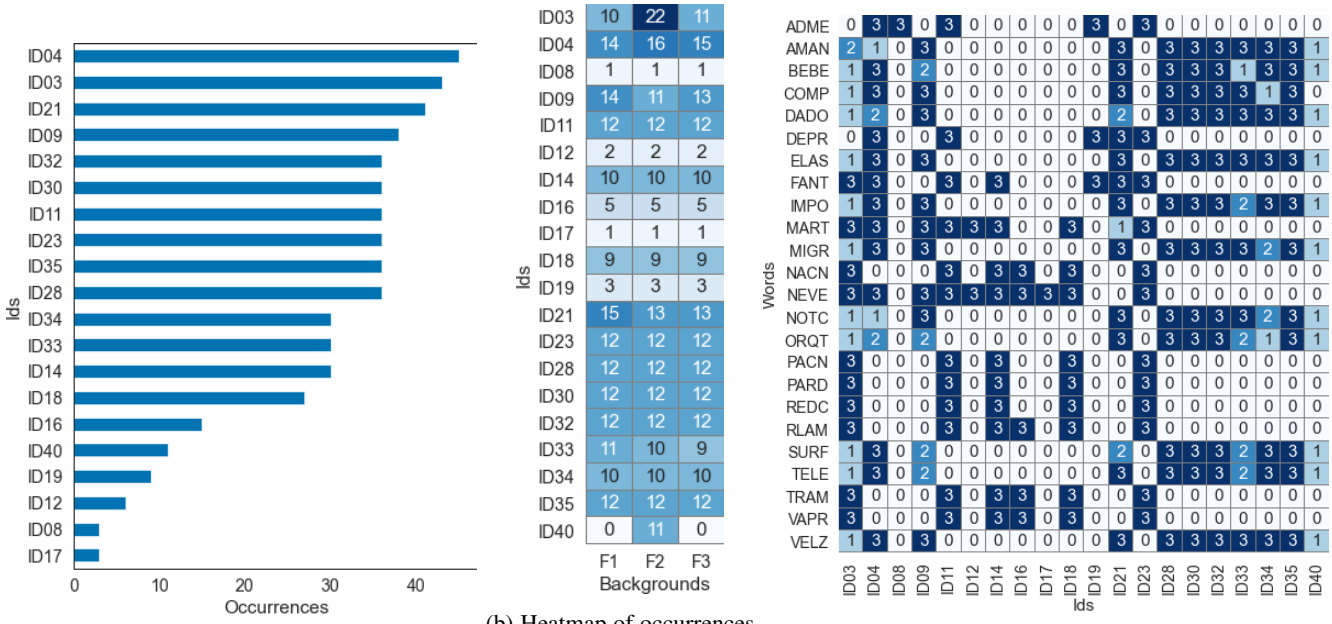
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(a) Occurrence of each word entry.

(b) Amount of frames per word entry.

Fig. 3: Word entries distribtion in the CEFET LIBRAS Database.



(a) Occurrence of each volunteer in the database.

(b) Heatmap of occurrences of each volunteer for each background.

(c) Heatmap of occurrences of each word for each volunteer.

Fig. 4: Volunteer distribution in the dataset.

applications in Libras. In this work, we describe in detail the CEFET Libras database, which is a challenging database with 547 videos and 24 dynamic signs. To serve as a baseline, we developed a computer vision-based system to recognize the gestures (the first one using the full database). In the proposed system, the feature mapping is performed by summing the squared residual video. This feature space (which we call S-Residue) can capture the Libras details, such as the face, hand configuration, as well as motion. The gestures in feature space were classified through K-NN and RF. In the k-fold cross-validation framework ($k = 3$), we achieved an average accuracy of 65.81%. Due to the complexity of the database, we consider it a satisfactory result.

REFERENCES

[1] (2010) Brazilian census of 2010. [Online]. Available: https://censo2010.ibge.gov.br/

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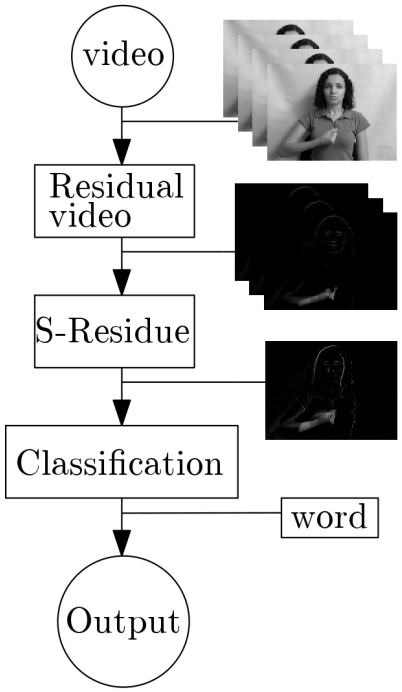


Fig. 5: Block diagram of the proposed method.

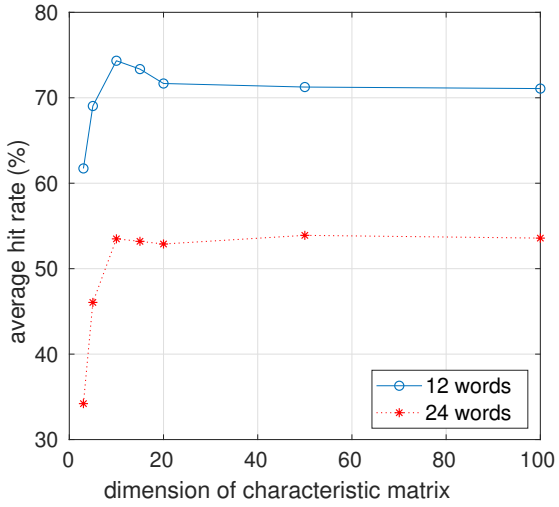


Fig. 6: K-NN ($k = 1$) Results. The blue curve was obtained by employing the same rationale applied in [15].

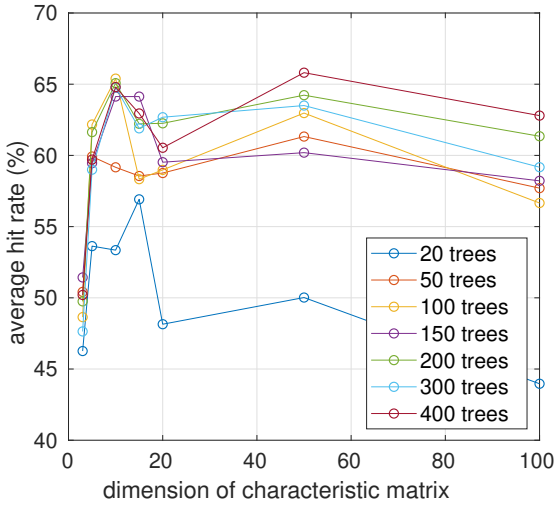


Fig. 7: Average accuracy when using RF varying the number of trees.

[2] J. Rocha, J. Lensk, T. Ferreira, and M. Ferreira, "Towards a tool to translate brazilian sign language (libras) to brazilian portuguese and improve communication with deaf," in *2020 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*, Dunedin, NZ, Aug. 2020, pp. 1–4.

[3] A. Xavier and C. Occhino, "Movement repetition in libras signs," in *17th Conference on Laboratory Phonology*, Vancouver, CA, July 2020, pp. 1–3.

[4] C. Harshith, K. R. Shastry, M. Ravindran, M. V. V. N. S. Srikanth, and N. Lakshmikhanth, "Survey on various gesture recognition techniques for interfacing machines based on ambient intelligence," *CoRR*, vol. 1, no. 2, pp. 31–42, Nov. 2010.

[5] K. P. de Souza, J. B. Dias, and H. Pistori, "Reconhecimento automático de gestos da lıngua brasileira de sinais utilizando visao computacional," in *III Workshop de Visão Computacional*, São José do Rio Preto, BR, Oct. 2007, pp. 1–6.

[6] Q. Munib, M. Habeeb, B. Takruri, and H. A. Al-Malik, "American sign language (asl) recognition based on hough transform and neural networks," *Expert systems with Applications*, vol. 32, no. 1, pp. 24–37, Jan. 2007.

[7] INES, "Dicionário da língua brasileira de sinais," 2005.

[8] A. T. Carneiro, P. C. Cortez, and R. C. Costa, "Reconhecimento de gestos da libras com classificadores neurais a partir dos momentos invariantes de hu," in *Anais do 1o Congresso Regional de Design de Interação - Interaction South America 09*, São Paulo, BR, Dec. 2009, pp. 190–195.

[9] C. R. de Souza and E. B. Pizzolato, "Sign language recognition with support vector machines and hidden conditional random fields: going from fingerspelling to natural articulated words," in *International Workshop on Machine Learning and Data Mining in Pattern Recognition*, New York, USA, July 2013, pp. 84–98.

[10] C. F. F. Costa Filho, R. S. d. Souza, J. R. d. Santos, B. L. d. Santos, and M. G. F. Costa, "A fully automatic method for recognizing hand configurations of brazilian sign language," *Research on Biomedical Engineering*, vol. 33, no. 1, pp. 78–89, Mar. 2017.

[11] S. Sharma, K. Kumar, and N. Singh, "Deep eigen space based asl recognition system," *IETE Journal of Research*, vol. 0, no. 0, pp. 1–11, July 2020.

[12] J. B. Dias, K. d. Souza, and H. Pistori, "Conjunto de treinamento para algoritmos de reconhecimento de libras," in *II Workshop de Visao Computacional, São Carlos*, São Carlos, BR, Oct. 2006, pp. 16–18.

[13] L. Yang, J. Chen, and W. Zhu, "Dynamic hand gesture recognition based on a leap motion controller and two-layer bidirectional recurrent neural network," *Sensors*, vol. 20, no. 7, pp. 2106–2123, Apr. 2020.

[14] E. Escobedo, L. Ramirez, and G. Camara, "Dynamic sign language recognition based on convolutional neural networks and texture maps," in *32nd Conference on Graphics, Patterns and Images*, Rio de Janeiro, BR, Oct. 2019, pp. 265–272.

[15] C. H. A. Monteiro, L. F. I. Pecoraro, A. T. Lacerda, A. R. Corbo, and G. M. Araujo, "Um sistema de baixo custo para reconhecimento de gestos em libras utilizando visão computacional," in *XXXIV Simpósio Brasileiro de Telecomunicações*, Manaus, BR, Sept. 2016, pp. 349–352.

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